S1. Occlusion resolver: evaluation contract

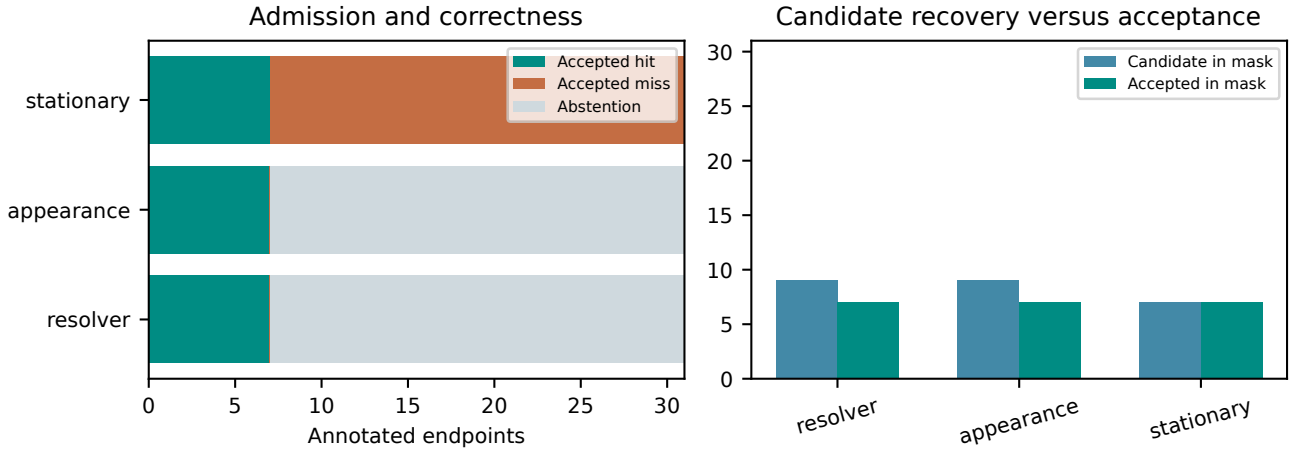
This supplement documents the executed causal prototype. It contains no audio–visual routing experiment. Initial query masks define object bounding rectangles; later target masks are used only after inference. The primary score is point membership inside a sparse target mask. It is not an amodal segmentation score, exact correspondence metric, or hidden-coordinate error.

Input. A decoded RGB video resized to 320×240 and the earliest query mask. All frames from the query through the last evaluation endpoint are processed in temporal order. No frame after an endpoint contributes to its stored state.

Output. A candidate, acceptance state, last supported point, evidence age, immutable appearance properties, and stationary/motion/context hypotheses. A view request is a symbolic action, not an executed camera command.

Population. The local directory contains 37 videos. Twenty-two have an eligible query and later target annotation, giving 31 endpoints. Some clips contribute several endpoints. This is a local subset, not an official TCOW benchmark reproduction.

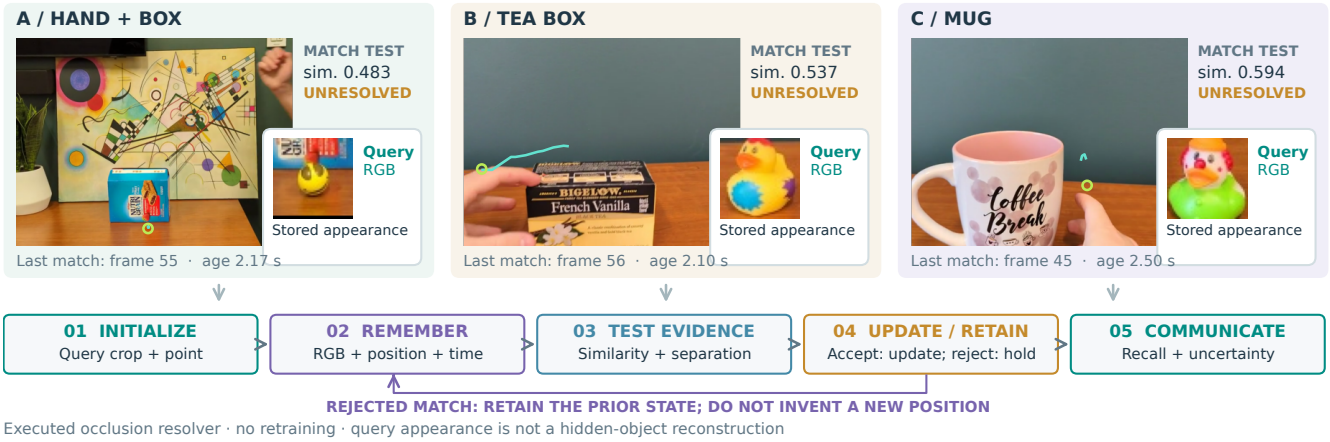
Reporting. Candidate hits, accepted hits, accepted misses and abstentions are reported together. A rejected correct proposal is not counted as a supported answer. Missing evidence is not counted as an exact location estimate.



The accepted-hit count is seven for both the resolver and appearance-only control; confirmation has no measured aggregate endpoint advantage. There are no accepted misses in this sample, but only seven accepted endpoints. This is not a population guarantee.

S2. Persistent object record

The identity record is the initial query RGB crop, grayscale template and HSV histogram. It is never overwritten during a loss interval or after reacquisition. Mean RGB and crop dimensions are directly measured properties and can include background. No material, affordance or semantic category estimator is implemented.



Field	Meaning
Query RGB	Immutable evidence from initialization
Last supported point	Latest accepted appearance candidate
Evidence age	Elapsed frame time since acceptance
Velocity	Smoothed displacement across accepted matches
Context coordinate	Local-flow transported hypothesis
Pending candidate	Consecutive-frame confirmation state

Important boundary. A supported match is still a model decision. A false accepted match can corrupt coordinates even though query appearance remains intact. The word supported therefore refers to the explicit appearance test, not ground-truth correctness.

S3. Candidate generation and acceptance

Three rescaled query templates are correlated globally with the current grayscale frame. At each scale only the strongest grayscale peak is considered. HSV histogram correlation reweights that proposal; it does not generate all possible color candidates. This is a specific weakness when a distractor dominates grayscale correlation.

$$z = 0.7\,\mathrm{NCC} + 0.3\,\mathrm{HC},$$
$$q = [z \geq 0.72] \wedge [\mathrm{NCC} \geq 0.55] \wedge [\mathrm{gap} \geq 0.025].$$

The gap is the strongest correlation minus the strongest value outside a neighborhood around the peak. It is not a posterior probability or conformal score. Histogram comparison uses 12 hue bins and eight saturation bins. The query bounding rectangle, not the exact mask silhouette, defines the template.

```
for frame in observed_prefix:
    candidate = best_of_three_scales(frame, immutable_query)
    qualifies = score >= .72 and ncc >= .55 and gap >= .025
    if qualifies and near_previous_pending_candidate:
        confirmation_count += 1
    else:
        confirmation_count = int(qualifies)
    accepted = qualifies and confirmation_count >= 2
    if accepted:
        update_last_position_and_velocity()
        reset_evidence_age()
    else:
        retain_last_supported_state()
    emit_observation_and_hypotheses_separately()
```

The appearance-only ablation removes confirmation and uses the same proposal scores and thresholds. Its endpoint totals tie the resolver; that negative result is retained in the paper.

S4. Hidden-state hypotheses

Three locations are exposed without collapsing them into a certain answer: the last supported point, a constant-velocity extrapolation capped at one second, and a local optical-flow transport. Median flow comes from a 20×20 neighborhood around the previous context coordinate. The flow is computed from consecutive observed frames with Farneback's algorithm.

What flow cannot establish. The tracked texture may belong to an occluder or background. No semantic container association is learned. The three hypotheses are alternative explanations, not independent measurements. Their probabilities are not calibrated and their display radius is heuristic.

How hypotheses are currently used. The API and visualization expose them for inspection and explanatory recall. They do not rerank the appearance proposal. Accordingly, endpoint gains cannot be attributed to motion-aware candidate selection. A future ablation must implement that decision dependency before claiming its benefit.

Time. The evaluator processes each frame at the source FPS. Age is counted in elapsed frame intervals. Motion velocity is updated only across accepted matches. A one-second extrapolation cap prevents arbitrarily large long-horizon displacement but does not make the predicted position correct.

View requests. Unresolved identity or location maps to a symbolic request for a view change. This release does not predict camera poses, execute a robot gaze command or measure whether a requested view resolves uncertainty. Such measurements require interventions or alternative observations.

S5. Baselines and information access

The expanded evaluation compares the resolver with appearance-only acceptance and a stationary initial center. The first shares every appearance component and isolates confirmation. The second always returns the initial center and never abstains. Both use the same query rectangle.

Archived learned trackers. The interactive scene explorer retains earlier CoTracker3 and CoTracker2 predictions on three selected clips. These models receive the prefix through an annotated scoring endpoint and may use future frames relative to an intermediate display. They are not causal baselines and were not newly evaluated on the expanded 22-clip set.

Initialization. Archived point trackers use a query point, while the current resolver uses a query-mask bounding rectangle. This additional appearance supervision is material. The website is a qualitative comparison, not a matched-supervision leaderboard.

Methods not run. TCOW, SAM 2, XMem and TAPIR are related work, not executed full-set baselines in this release. No table should suggest otherwise. A strong paper requires their evaluation or an appropriate task-matched subset, with timing, initialization and future-frame access controlled.

Scope of the ablation. Confirmation and non-confirmation tie on endpoint counts. Differences in intermediate acceptance could be useful or harmful; sparse endpoint masks cannot determine which. Dense visibility and identity annotations would be needed for a temporal claim.

S6. Expanded per-clip endpoint audit

Clip	Endpoints	Candidate hits	Accepted hits	Accepted misses
bagduck1_move	1	0	0	0
chips1_oof	2	1	0	0
confuse1	1	1	1	0
confuse3	1	0	0	0
handball1_wave	1	0	0	0
handball2_boxoccl	1	0	0	0
handball3_occlpick	1	0	0	0
lightball4_occl1x	3	0	0	0
lightball5_occl2x	2	0	0	0
lightball6_occl4x	3	0	0	0
mugduck2_reveal	1	0	0	0
mugduck3_reveal	1	1	0	0
mugduck5_stay	1	0	0	0
mugduck6_shuffle	1	0	0	0
mugduck7_shuffle	2	0	0	0
plantcupball2	1	0	0	0
pump1_scan	2	1	1	0
pumpcookie1_reveal	1	0	0	0
teaduck1	1	1	1	0
teaduck2	1	1	1	0
teaduck3_reveal	1	1	1	0
teaduck4_reveal	2	2	2	0

Repeated endpoints from a clip are clustered observations. The table includes every eligible clip, not only the three shown on the website. Candidate membership is scored even when the resolver rejects it. The complete machine-readable table also contains every baseline row.

S7. Implementation verification

Four tests execute: immutable memory and retained coordinates during loss; two-frame confirmation; deterministic prefix behaviour; finite separate hypotheses and an unresolved-state view request. All pass. Synthetic test images exercise code invariants only and are not visual benchmark results.

Replay integrity. A query-anchor byte hash is checked before and after every evaluated clip. The evaluator checks that the immutable RGB crop has not changed. This does not certify correct localization: it certifies the intended memory policy.

No endpoint leakage. The inference loop reads RGB frames only. Target masks are opened in the following scoring loop. Query annotations are permitted initialization. Offline tracker traces are used only by the qualitative website and are not read by the resolver.

Known limits. Global template search is sensitive to texture, crop background and viewpoint. Correlation is not identity. The current rejection policy gives low answer coverage. Testing code invariants must not be described as proving perception robustness.

S8. Evidence and claim ledger

Claim	Status
Immutable appearance record	Implemented and unit-tested
Causal frame processing	Executed on 22 eligible clips
Two-frame confirmation	Implemented; no endpoint gain over appearance control
Accepted target-mask hits	7 of 31 endpoints
Accepted out-of-mask predictions	0 of 31 endpoints, with 24 abstentions
Exact hidden position	Not measured or established
Semantic object properties	Not inferred; stored RGB and crop dimensions only
Container-aware tracking	Not implemented; local flow is a hypothesis
Robot conversation	Deterministic state-to-text template
Ameca control or transfer	Not executed
Novelty or state-of-the-art superiority	Not established

Research direction. A useful next mechanism must change a measured decision: containment inference could propagate a target with a moved container; an active-view policy could reduce ambiguity; learned re-identification could recover appearance changes. None should be presented as an achieved capability before it is implemented and compared.

S9. Reproduction and machine interface

```
python -m pip install -r requirements.txt
python code/test_resolver.py
python code/evaluate.py --data /path/to/rubric_office --output results
python code/report_assets.py
python memory_figures.py
```

The data must be acquired from the TCOW project with its original terms and attribution. The ZIP includes the three demonstration videos already used by the site, not the entire upstream dataset. Scripts use deterministic matching; no training seed or learned checkpoint is needed for the new resolver.

Records. Each trace contains frame, candidate, accepted flag, state, last supported coordinate, evidence age, three hypotheses, a heuristic radius, NCC, color similarity, appearance score, ambiguity gap and query properties. The current code additionally emits a symbolic view request or continued-observation action.

Units. Positions and radii use the 320×240 evaluation raster. Velocity uses pixels per second. Masks are evaluated after scaling and rounding coordinates to annotation resolution. These are not metric robot-space coordinates.

Case pages. The next eleven pages show two clips each. Query RGB appears on the left; the last scored endpoint appears on the right with a green evaluation-mask outline. The outline is not supplied to inference. Tables include all scored endpoints for the resolver. Appearance and stationary records are in the JSON. No hidden-location ground truth is invented.

S10. Per-clip records

bagduck1_move




Query frame 20; displayed endpoint 230. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
230	0	0	6.57

chips1_oof




Query frame 40; displayed endpoint 280. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
160	1	0	3.20
280	0	0	2.47

S11. Per-clip records

confuse1



Query frame 20; displayed endpoint 136. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
136	1	1	0.00

confuse3



Query frame 15; displayed endpoint 73. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
73	0	0	1.17

S12. Per-clip records

handball1_wave



Query frame 5; displayed endpoint 137. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
137	0	0	4.20

handball2_boxoccl



Query frame 30; displayed endpoint 210. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
210	0	0	5.17

S13. Per-clip records

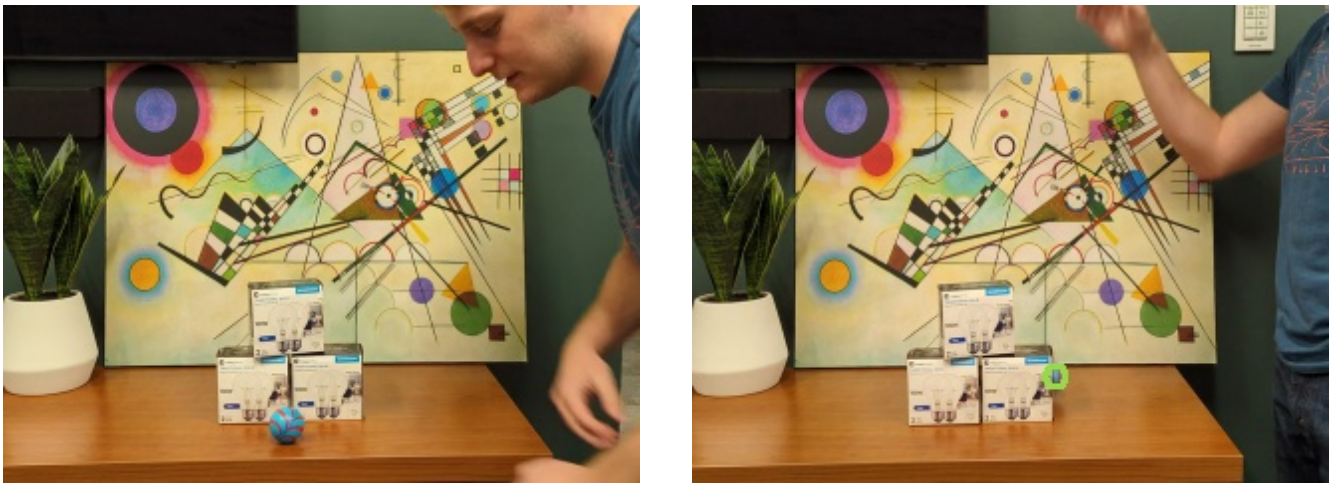
handball3_occlpick



Query frame 20; displayed endpoint 190. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
190	0	0	5.47

lightball4_occl1x

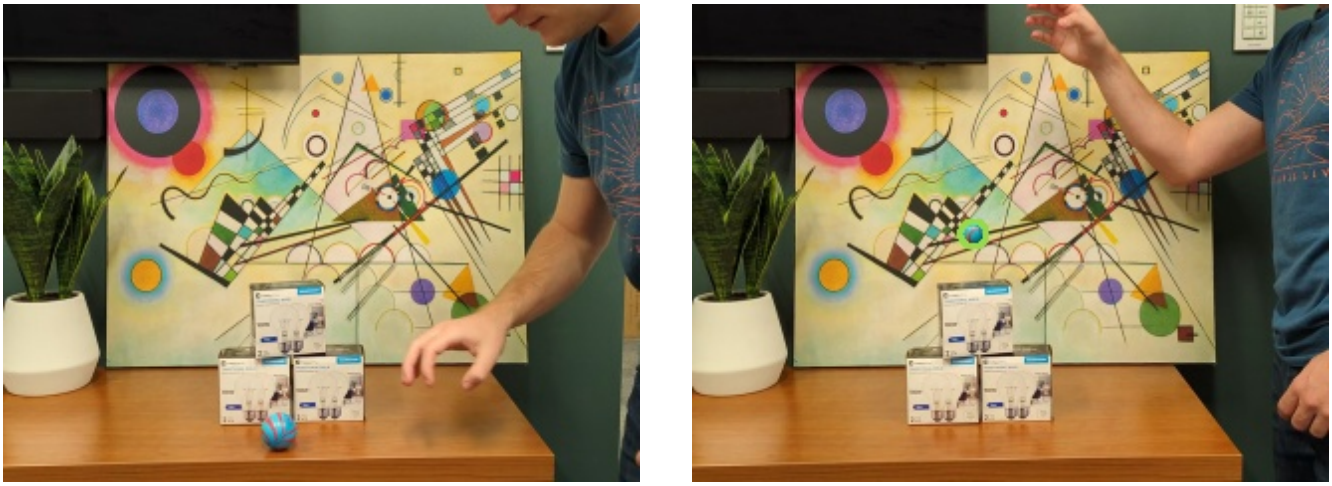


Query frame 40; displayed endpoint 340. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
190	0	0	0.78
312	0	0	2.82
340	0	0	3.28

S14. Per-clip records

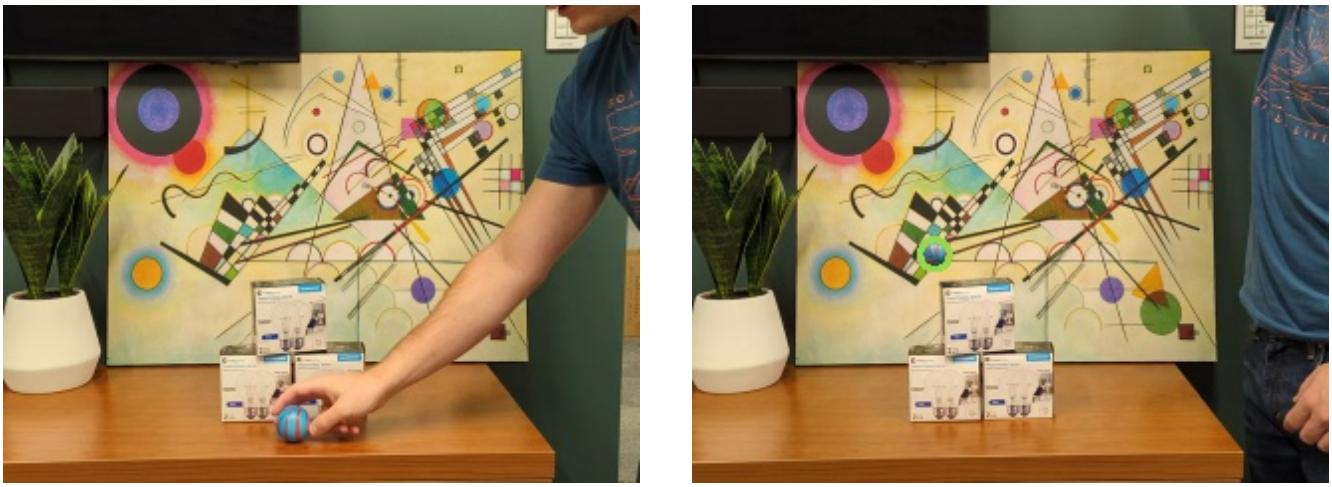
lightball5_occl2x



Query frame 30; displayed endpoint 236. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
170	0	0	1.85
236	0	0	2.95

lightball6_occl4x



Query frame 15; displayed endpoint 290. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
175	0	0	2.13
226	0	0	2.98
290	0	0	4.05

S15. Per-clip records

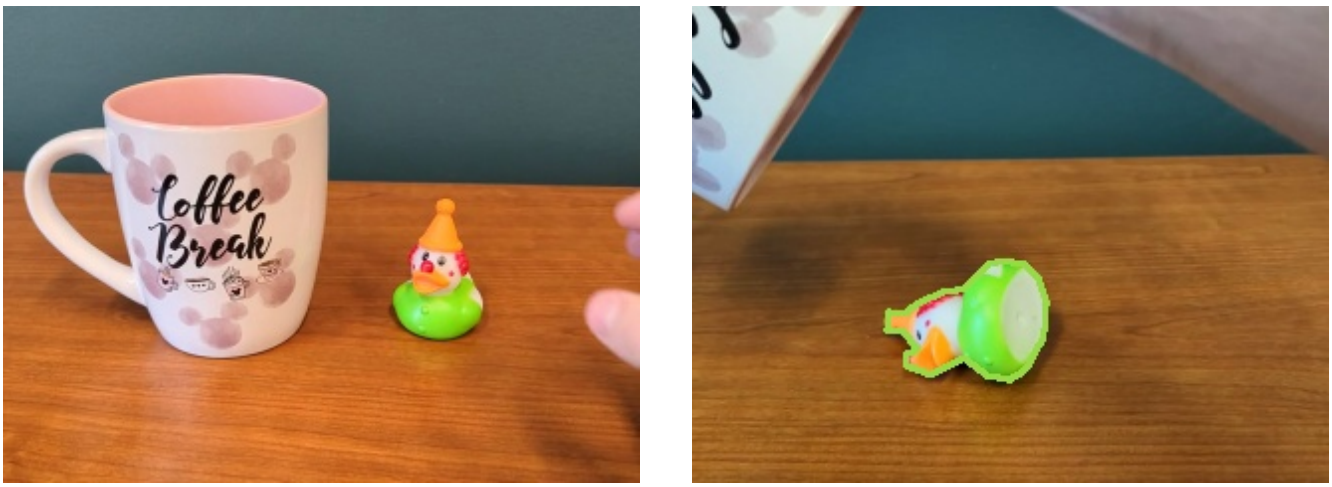
mugduck2_reveal



Query frame 20; displayed endpoint 200. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
200	0	0	5.17

mugduck3_reveal

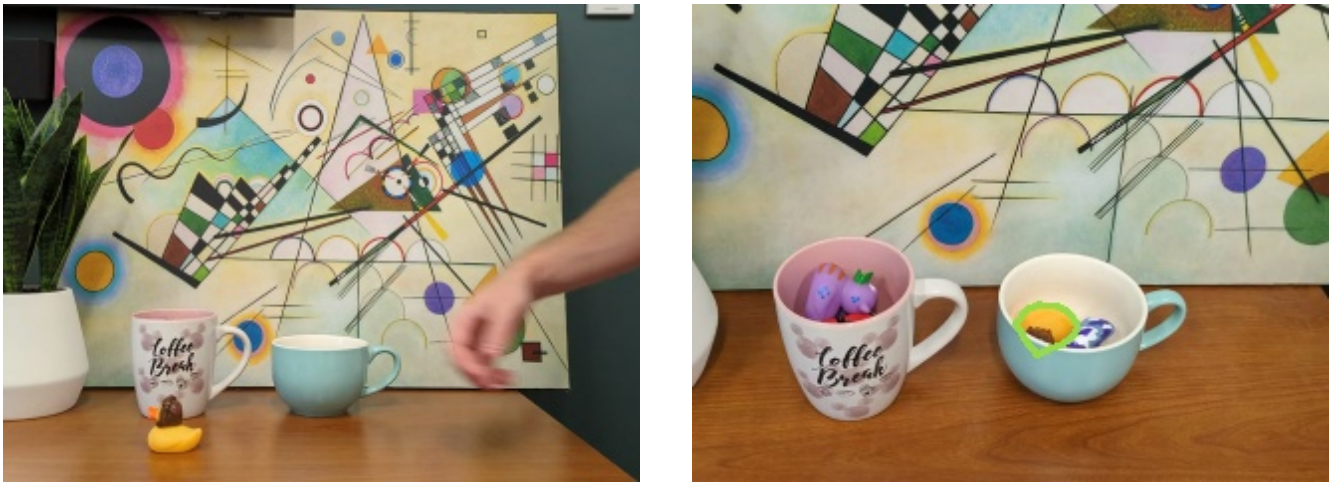


Query frame 15; displayed endpoint 195. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
195	1	0	5.77

S16. Per-clip records

mugduck5_stay



Query frame 20; displayed endpoint 140. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
140	0	0	3.70

mugduck6_shuffle



Query frame 30; displayed endpoint 210. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
210	0	0	5.47

S17. Per-clip records

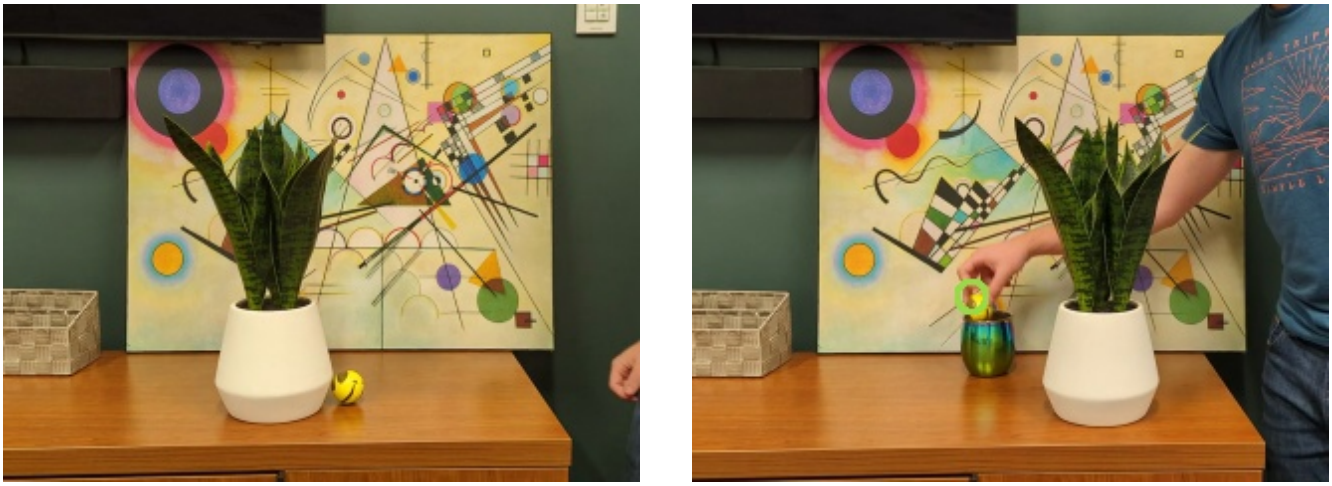
mugduck7_shuffle



Query frame 30; displayed endpoint 210. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
190	0	0	4.40
210	0	0	5.07

plantcupball2



Query frame 25; displayed endpoint 265. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
265	0	0	6.67

S18. Per-clip records

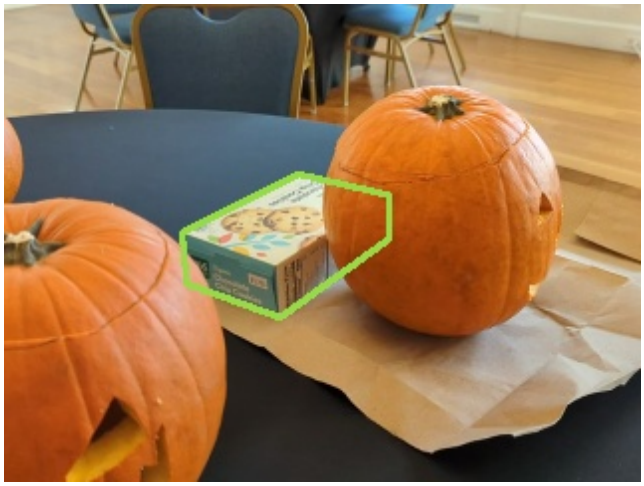
pump1_scan




Query frame 30; displayed endpoint 270. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
150	0	0	0.63
270	1	1	0.00

pumpcookie1_reveal

Query frame 20; displayed endpoint 200. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
200	0	0	5.53

S19. Per-clip records

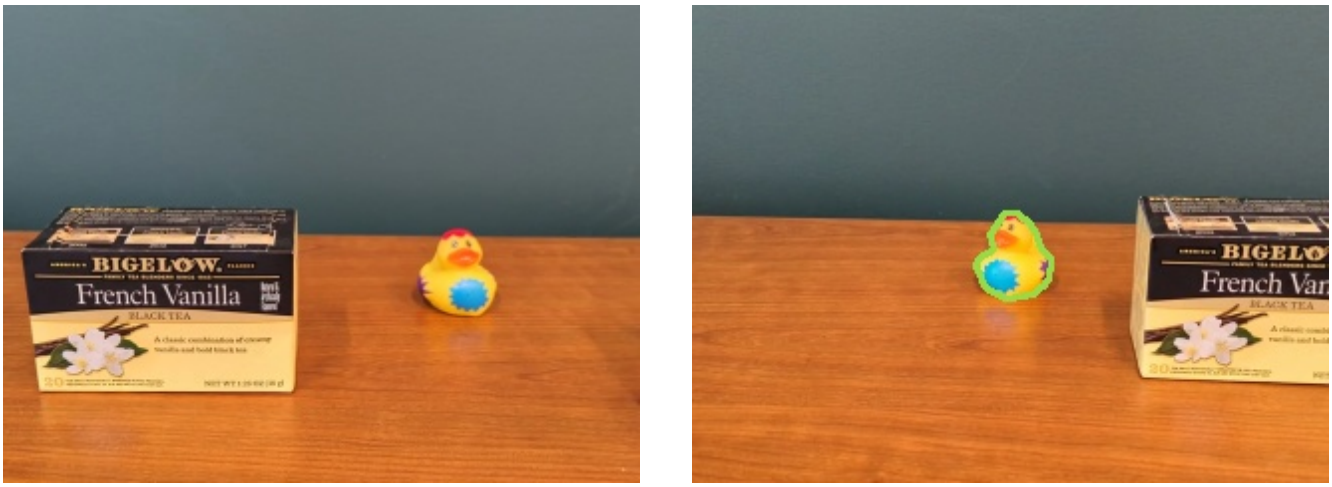
teaduck1



Query frame 50; displayed endpoint 170. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
170	1	1	0.00

teaduck2



Query frame 15; displayed endpoint 195. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
195	1	1	0.00

S20. Per-clip records

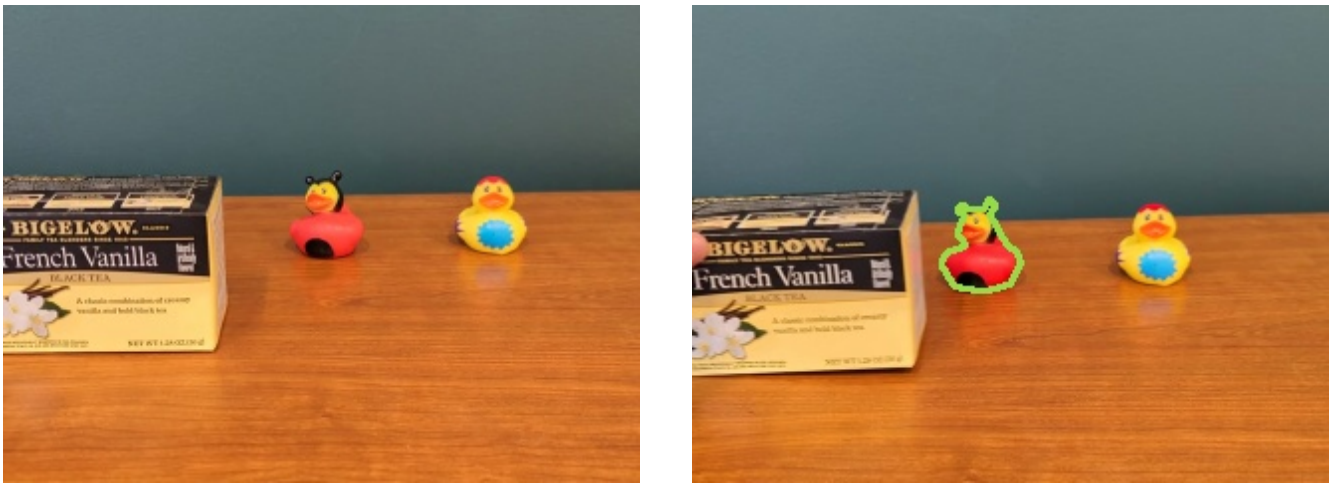
teaduck3_reveal



Query frame 15; displayed endpoint 195. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
195	1	1	0.00

teaduck4_reveal



Query frame 15; displayed endpoint 255. Green outline: evaluation target mask, not model input.

Frame	Candidate inside	Accepted	Evidence age (s)
160	1	1	0.00
255	1	1	0.00